

AI-Based Stepwise Algebra Evaluation Report

Benchmark Evaluation Results: Teacher Ground Truth vs AI System

1. Question-Level Metrics

Metric	Value
MAE	0.9643
Pearson Correlation (r)	0.5332
Exact Match Accuracy	64.29%
Exact Match F1	0.7826
Mean Bias Error (MBE)	-0.8214
RMSE	2.2835
R ² Score	-0.1124

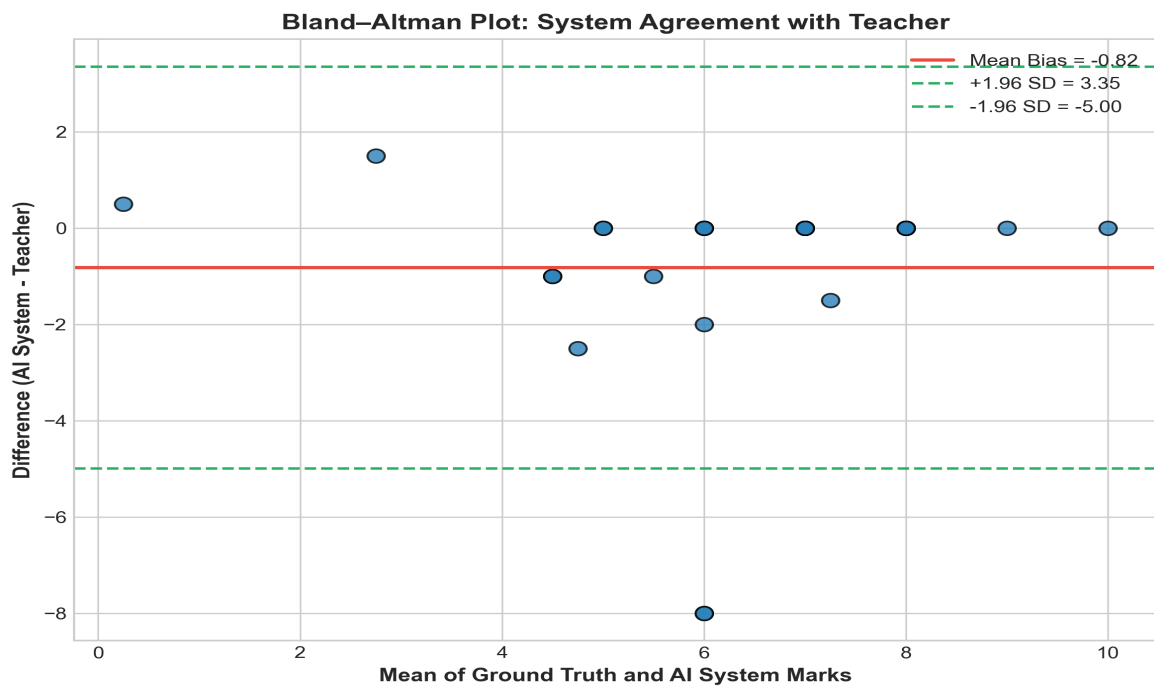
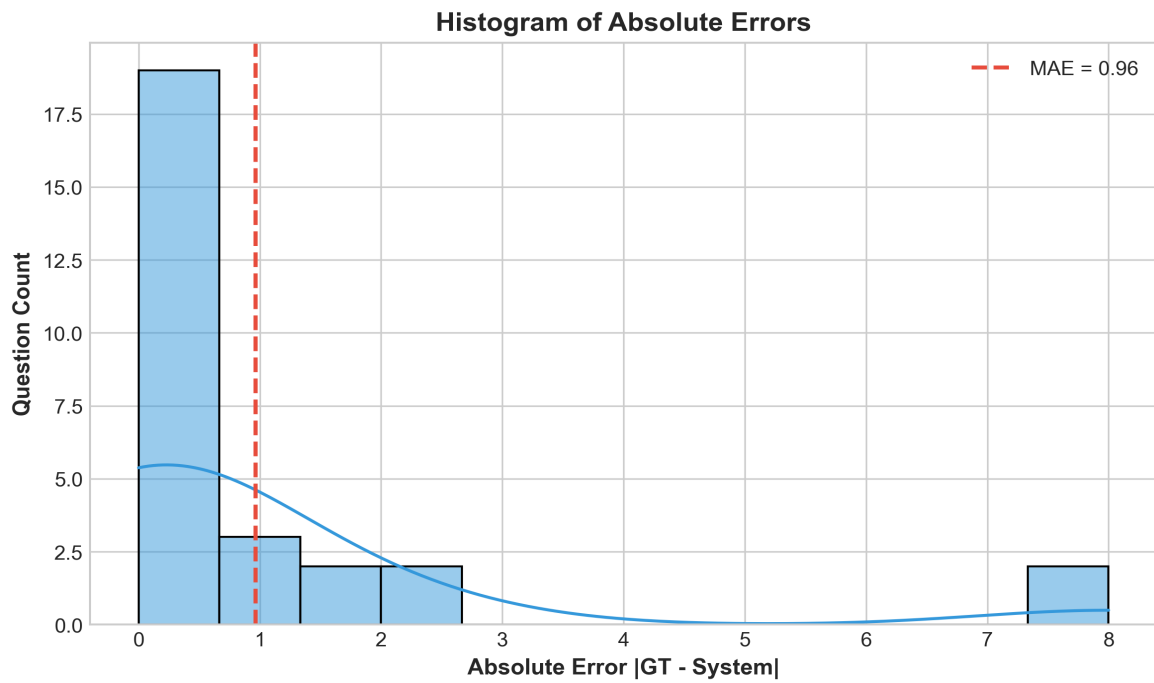
2. Step-Level Evaluation (Validity Classification)

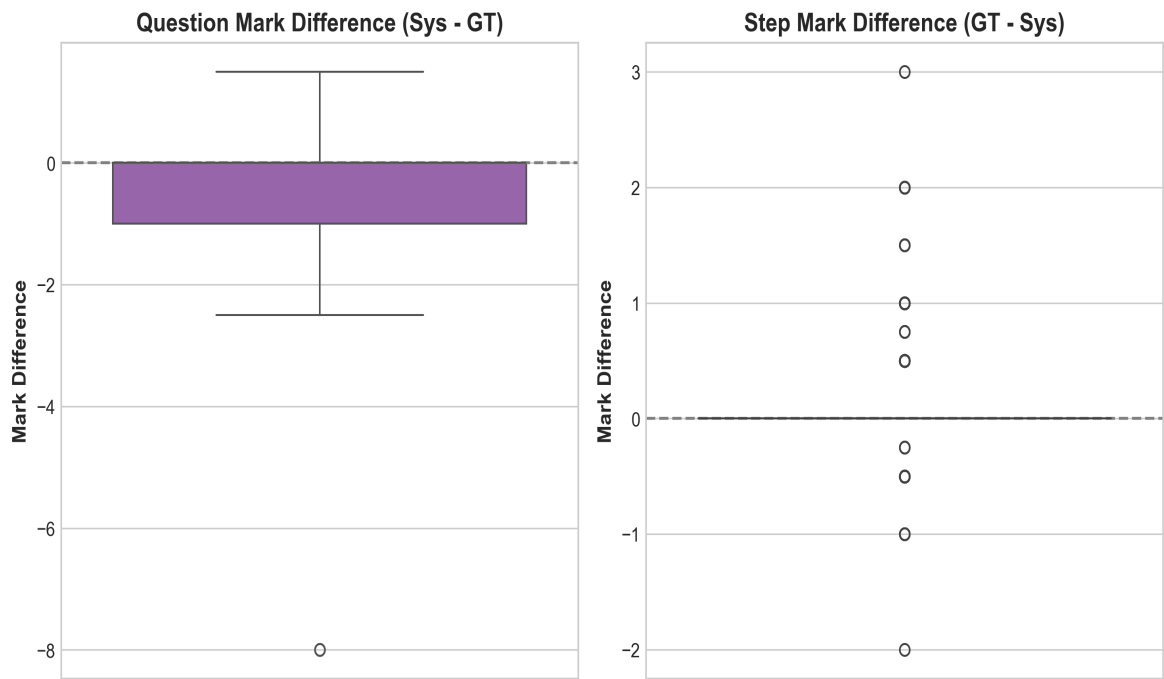
Metric	Value
Macro F1	0.4351
Weighted F1	0.8936
Micro F1	0.8814
F1 (correct)	0.9350
F1 (partially_correct)	0.0000
F1 (incorrect)	0.3704
Cohen's κ	0.2729
Accuracy	88.14%
Precision	0.9139
Recall	0.8814

3. Step Marking Evaluation

Metric	Value
Step MAE	0.2599
Step RMSE	0.5761
Pearson r	0.5562
Mean Difference	-0.1299

4. Visualizations





Step Validity Confusion Matrix				
Ground Truth (Teacher)	correct	incorrect	partially_correct	
	151	14	1	
	3	5	0	
	3	0	0	
		correct	incorrect	partially_correct
		Predicted (AI System)		

